

MCQS

MTH100 midterm preparation file

ZB

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

In the name of God , Most
Gracious, Most merciful



MTH100

MCQS

1. Which of the following is an Arithmetic Sequence?

- a) 35, x , $x+2$, $x+4$, ...
- b) 35, x , $x+2$, $x+4$, $x+6$, ...
- c) 24, 35, x , $x+5$, $x+10$, ...
- d) 24, 35, x , $x+5$, $x+10$, $x+15$, ...

Answer: b) 35, x , $x+2$, $x+4$, $x+6$, ...

2. The midpoint of line segment AB with coordinates A(5, 8) and B(3, 6) is:

- a) (1, 1)
- b) (4, 7)
- c) (6.5, 4.5)
- d) (-1.5, -1.5)

Answer: b) (4, 7)

3. For the arithmetic sequence , which of the following is the best option for 'd'?

- a) First term
- b) Last term
- c) Constant
- d) Common difference

Answer: d) Common difference

4. What is the common ratio 'r' of the geometric sequence 5, 10, 20, 40, ...?

- a) -5
- b) $\frac{1}{2}$
- c) 2
- d) 5

Answer: c) 2

5. The determinant of matrix is:

- a) 12
- b) 10
- c) 14
- d) 8

Answer: b) 10

21. What defines an arithmetic sequence?

- a) Each term is the square of its position.
- b) Each term is obtained by adding a constant value to the previous term.
- c) Each term is obtained by multiplying the previous term by a constant.
- d) Each term is a random number.

Answer: b) Each term is obtained by adding a constant value to the previous term.

22. Which of the following statements is true for a geometric sequence?

- a) The difference between consecutive terms is constant.
- b) The ratio between consecutive terms is constant.
- c) The sum of consecutive terms is constant.
- d) The terms decrease linearly.

Answer: b) The ratio between consecutive terms is constant.

23. In an arithmetic sequence, what is the term “common difference” used to describe?

- a) The last term in the sequence.
- b) The value that is subtracted from each term to get the next term.
- c) The constant value added to each term to get the next term.
- d) The ratio between consecutive terms.

Answer: c) The constant value added to each term to get the next term.

24. What is the name of a sequence in which each term approaches a fixed number as the sequence progresses?

- a) Divergent sequence**
- b) Arithmetic sequence**
- c) Geometric sequence**
- d) Convergent sequence**

Answer: d) Convergent sequence

25. What is a series in mathematics?

- a) A sequence of random numbers.**
- b) The sum of the terms of a sequence.**
- c) A sequence where each term is squared.**
- d) A collection of unrelated numbers.**

Answer: b) The sum of the terms of a sequence.

26. What is the first term of a sequence called?

- a) Initial difference
- b) Constant ratio
- c) First element
- d) First term

Answer: d) First term

27. Which of the following best describes a divergent sequence?

- a) A sequence that approaches zero as it progresses.
- b) A sequence that approaches a fixed finite value.
- c) A sequence that grows without bound or oscillates indefinitely.
- d) A sequence that remains constant throughout.

Answer: c) A sequence that grows without bound or oscillates indefinitely.

28. What type of sequence is formed when the difference between consecutive terms is not constant, but the ratio is constant?

- a) Arithmetic sequence
- b) Geometric sequence
- c) Harmonic sequence
- d) Random sequence

Answer: b) Geometric sequence

29. In a finite sequence, what does the term "n" typically represent?

- a) The sum of the sequence.
- b) The position of the last term.
- c) The value of the first term.
- d) The common ratio.

Answer: b) The position of the last term.

30. What is the sum of all terms in an infinite geometric series called if the series converges?

- a) Arithmetic sum
- b) Geometric limit
- c) Finite sum
- d) Infinite sum

Answer: d) Infinite sum

31. What is the defining property of a finite sequence?

- a) It has an infinite number of terms.
- b) It contains a fixed number of terms.
- c) The terms always decrease.
- d) The terms always increase.

Answer: b) It contains a fixed number of terms.

32. Which type of sequence is formed when the reciprocal of each term creates an arithmetic sequence?

- a) Arithmetic sequence
- b) Geometric sequence
- c) Harmonic sequence
- d) Fibonacci sequence

Answer: c) Harmonic sequence

33. What is the general term of a sequence often used to represent?

- a) The difference between terms.
- b) The formula for any term in the sequence.
- c) The sum of all terms in the sequence.
- d) The total number of terms in the sequence.

Answer: b) The formula for any term in the sequence.

34. What do we call a sequence in which each term is the sum of the two preceding terms?

- a) Arithmetic sequence**
- b) Geometric sequence**
- c) Fibonacci sequence**
- d) Divergent sequence**

Answer: c) Fibonacci sequence

35. Which of the following best defines a series?

- a) A list of terms arranged in a particular order.**
- b) The result of multiplying terms in a sequence.**
- c) The result of adding terms in a sequence.**
- d) A sequence with constant differences.**

Answer: c) The result of adding terms in a sequence.

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36. What is the common ratio in a geometric sequence?

- a) The ratio of the first term to the second term.**
- b) The constant value obtained by dividing any term by its preceding term.**
- c) The difference between two consecutive terms.**
- d) The product of all terms in the sequence.**

Answer: b) The constant value obtained by dividing any term by its preceding term.

37. What is a sequence that alternates between positive and negative terms called?

- a) Harmonic sequence**
- b) Alternating sequence**
- c) Arithmetic sequence**
- d) Periodic sequence**

Answer: b) Alternating sequence

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38. What do you call a sequence that has no specific pattern?

- a) Arithmetic sequence**
- b) Geometric sequence**
- c) Random sequence**
- d) Divergent sequence**

Answer: c) Random sequence

39. What is the difference between a sequence and a series?

- a) A sequence is a list of numbers, while a series is the sum of those numbers.**
- b) A sequence is infinite, while a series is finite.**
- c) A sequence involves multiplication, while a series involves addition.**
- d) A sequence always converges, while a series always diverges.**

Answer: a) A sequence is a list of numbers, while a series is the sum of those numbers.

40. What is the sum of the terms in an arithmetic sequence called?

- a) Geometric series**
- b) Arithmetic mean**
- c) Arithmetic series**
- d) Arithmetic progression**

Answer: c) Arithmetic series

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41. Which of the following is true for an infinite geometric series to converge?

- a) The first term must be zero.
- b) The common ratio must be greater than 1.
- c) The common ratio must be less than 1 in absolute value.
- d) The series must have only positive terms.

Answer: c) The common ratio must be less than 1 in absolute value.

42. What is the process of finding the sum of a sequence called?

- a) Summation
- b) Integration
- c) Progression
- d) Product

Answer: a) Summation

43. In an arithmetic sequence, if the common difference is negative, what happens to the terms?

- a) They remain constant.
- b) They increase.
- c) They decrease.
- d) They alternate in sign.

Answer: c) They decrease.

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44. What does the term "convergent sequence" mean?

- a) A sequence that approaches a finite value as the number of terms increases.**
- b) A sequence that increases indefinitely.**
- c) A sequence that oscillates between two values.**
- d) A sequence where terms alternate in sign.**

Answer: a) A sequence that approaches a finite value as the number of terms increases.

45. What is the primary difference between arithmetic and geometric sequences?

- a) Arithmetic sequences have a common ratio, while geometric sequences have a common difference.**
- b) Arithmetic sequences have a constant difference between terms, while geometric sequences have a constant ratio between terms.**
- c) Arithmetic sequences are always finite, while geometric sequences are always infinite.**
- d) Arithmetic sequences increase, while geometric sequences decrease.**

Answer: b) Arithmetic sequences have a constant difference between terms, while geometric sequences have a constant ratio between terms.

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1. Basic Algebra

Q: Solve for x in the equation .

- a) 1
- b) 3
- c) 9
- d) 11

Answer: b) 3

2. Linear Equations

Q: The slope of the line passing through points (2, 4) and (6, 8) is:

- a) 1
- b) 2
- c) 0.5
- d) 4

Answer: a) 1

3. Quadratic Equations

Q: What is the discriminant of the quadratic equation ?

- a) 0
- b) 4
- c) -16
- d) 8

Answer: a) 0

4. Probability

Q: If a coin is tossed, what is the probability of getting heads?

- a) $\frac{1}{3}$
- b) $\frac{1}{2}$
- c) $\frac{1}{4}$
- d) $\frac{2}{3}$

Answer: b) $\frac{1}{2}$

5. Basic Geometry

Q: The area of a triangle with a base of 6 units and a height of 4 units is:

- a) 24 sq. units
- b) 12 sq. units
- c) 10 sq. units
- d) 20 sq. units

Answer: b) 12 sq. units

1. Which of the following is a prime number?

- a) 4
- b) 9
- c) 17
- d) 20

Answer: c) 17

2. What is the value of ?

- a) 17
- b) 18
- c) 19
- d) 20

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Passing criteria of this course

Letter Grade	Grade Points	Equivalent Percentage
A+	4.00	90-100
A	4.00	85-89
A-	3.66 - 3.99	80-84
B+	3.33 - 3.65	75-79
B	3.00 - 3.32	71-74
B-	2.66 - 2.99	68-70
C	2.00 - 2.65	61-67
D	1.00 - 1.99	50-60
F	0.00	00-49

Passing Criteria for a Course:

The passing criteria is defined in such a way that it ensures the student shall be consistent in his studies throughout the semester.

Therefore, for passing a course/subject, student shall fulfill the following:

- Secure minimum **20%** score in Formative Assessments/Mid terms
- Secure minimum **20%** score in Final Term Examinations.
- Secure at least **40%** marks in aggregate while fulfilling the above requirements

Course Selection and Credit Hours

Q: How do I select courses when the course selection is open?

A: Follow these steps:

- 1. Check the credit hours allowed by your university (e.g., 21, 18, or 15 credit hours).**
- 2. Each subject typically has 3 credit hours.**
- 3. Divide the total allowed credit hours by 3 to determine how many subjects you can select.**

Examples:

21 credit hours = 7 subjects

15 credit hours = 5 subjects

Remember, the university may allow different credit hours for each student, so check your specific allowance.

By following these steps, you can make informed decisions during course selection and manage your credit hours effectively.

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Advice

Hmaesha relax ho ke parha kryn panic mat hua kryn apky parents ky bohat khawab hoty hn wo pura krny ki puri koshish kea kryn kbi b ksi pe depend na kea kryn apko bs ak insan success kr saqta ha or wo insan ap khud hn apky elawa koi nahe ap bs koshish kryn Allah pak pe strong yakeen rakha kryn or mehnat krke sb Allah ke hawaly kr dea kryn everything is possible be brave be strong stay blessed

Hum insan hamesha moat se darty or bhagty hn or moat se bachny ki koshish krty hn jabke moat ny ana hi ana ha hum

moat se nahe bach saqty humy jahanam se bachny ki koshish
krni chaheay hum jahanam se bach saqty hn
Insan ko 3 chezo se dar lagta ha Moat Risk/Dolat Ezat Shohrat
Fame

Moat

**Humy pta hona chaheay Moat tab ani ha jab Allah
chahy us se pehly puri dunya bi ak taraf ho ke apko
marna chahy to apka kuch nahe beggar saqti**

Risk/Dolat

**Risk Dolat sb Allah pak ke hath m ha puri dunya ak
taraf b ho jay na apsy apka risk cheen saqti ha na de
saqti ha Ye sirf ALLAH pak ky hath m ha**

Izat Zilat shohrat Fame

**Izat Zilat Sirf Allah de saqty hn puri dunya mel ke b
apko zra brabr b damage nahe kr saqti Agr ALLAH**

**Apko izat dena chahay to puri dunya ak taraf ho ke
b apko 1% b nuksan nahe pohancha saqti**

Phr Dar ks bat ka ????????????????

**Be brave be strong Just put your trust To ALLAH
Hamesha Confident or himat se raho kbi ksi k samne
mat jhuko puri dunya mel k b apka kuch b nahe
beggar saqti**

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MY REQUEST FROM ALL OF YOU

ZB Request from All of you My family

Mjy ap sbki help or support ki zarort ha or wo ye k hum sb Mel kr Allah ka Quran ki Urdu translation logo tak pohanchy or Quran ko samjna asaaan kryn dosro k leay sb tak Allah ka Quran pohanchy it's my campaign hum log 70+ age k ho jaty hn phr b hmy namaz tak ki translation nahe pta hoti k hum Allah pak se

Kya Dua kr rhy hn so hum youngster's ko Mel kr puri takat or energy ke sath Allah ka Quran spread krna ha

Hum puri Koshish kr rhy hn k hum sb Mel k Quran ki translation or most important topics ko maximum share kry with translation hmara maksad Quran ko spread krna ha be a part of us

m apni pocket se ye sb kr rha ho Allah ka Quran spread krny k leay apne or mene sbny Marna ha ak din to q na Allah k leay Kuch kryn apni energies apni power ko bajay negative use krny k Allah k leay invest kryn khud ko be a part of us

Agr m 23 ki age m apny sare sources use kr rha ho pocket se heavy amounts give away kr rha ho Quran spread krny k leay to ap just share to kr saqty hn itna e krdyn

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Agr ap mujsy contact na kr sky too b ap jo b hn jaha bi hn waha Allah ka quran pheelay zada se zada with urdu translation jitni himat ha utna share kryn

..... **MY Family**

**ust Relax and focus! Exams are not difficult. Put in your
100% effort and trust in Allah.**

..... **AL-Quran**

**"And indeed, with hardship comes ease." (Quran 94:5)
Remember, I'm here to support you! Stay brave and strong!"**

May Allah bless you